CHROMATIC NUMBER AND SUBGRAPHS OF CAYLEY GRAPHS

László Babai
Eötvös L. University
Budapest, HUNGARY

Abstract

Results, problems and conjectures concerning the chromatic number and subgraphs of finite and infinite Cayley graphs are presented. We prove that every group has a Cayley graph of chromatic number $\leq \omega$; for solvable groups the minimum chromatic number is ≤ 3. The Cayley graph of an irreducible group presentation contains neither $K_{3,5}$ nor $\bar{K}_4$ as a subgraph. Every group has a Cayley graph containing neither $\bar{K}_4$ nor $K_{5,17}$. We mention that every graph is a induced subgraph of some Cayley graph of any sufficiently large group.

1. Introduction. The Cayley graph $X = X(G, H)$ of the group G with respect to the generating set H is defined by

$$V(X) = G,$$

$$E(X) = \{[g, hg] : g \in G, h \in H\}.$$

It is almost a century since Cayley graphs have been used to visualize group structure, and Burnside and Maschke have already investigated their imbeddability on surfaces. By now, a powerful theory of imbeddings of Cayley graphs is developed with several applications (see COXETER, MOSER [7] and A. T. WHITE [23] for details). Some other problems, such as that of the connectivity and the 1-factors, are settled generally for graphs with transitive automorphism groups (the connectivity by WATKINS [21], MADER [17, 18] and JUNG

[13]; the 1-factors by LITTLE, GRANT and HOLTON [15]; LOVÁSZ has shown that vertex-transitive graphs with an even number of vertices are even bicritical (personal communication). For both problems, see Chapter 12 of the excellent book [16] by L. LOVÁSZ.)

Almost nothing seems to be known, however, about the relationship of Cayley graphs to a number of classical graph theoretic notions, such as chromatic number, Hamilton paths, containment of certain subgraphs, etc. The present note is by no means intended to provide a systematic treatment of the subject. It is the author's hope to stimulate research in this area by presenting a few results and mentioning numerous problems and conjectures.

Concerning Hamilton circuits, we must rest satisfied with conjecturing that each Cayley graph has a Hamilton circuit, and each connected graph with a transitive automorphism group has a Hamilton path. The former can be easily proven for abelian groups, as remarked by J. PELIKÁN.

In this paper, we investigate the minimum chromatic number of Cayley graphs, and the problem as to whether a subgraph necessarily occurs in any Cayley graph of a group.

Acknowledgement: the author is indebted to P. Frankl for his valuable suggestions.

2. Preliminaries.

We deal with undirected graphs without loops or multiple edges. In order that the Cayley graph $X(G, H)$ be loopless, we always suppose $1 \notin H$. (1 denotes the unit of G.) For basic information on Cayley graphs see [19, 22]. We let $\mathrm{ch}\ X$ stand for the chromatic number of the graph X; K_m denotes the complete m-graph and $K_{m,n}$ the complete (m, n) bipartite graph.

Throughout, G denotes a (finite or infinite) group. For $H \subseteq G$, H is an irreducible generating set, if $\langle H \setminus \{h\} \rangle \neq G$ for

any $h \in H$.

H is a quasi-irreducible generating set, if $\langle H \rangle = G$ and H can be well-ordered such that no member of H is contained in the subgroup generated by its predecessors.

THEOREM 2.1. *If H is a quasi-irreducible generating set of G then $\mathrm{ch}\, X(G, H)$ is at most countable.*

The proof follows in Section 3.

CONJECTURE 2.2. *There is a finite m such that for H a quasi-irreducible (irreducible, resp.) generating set of G, $\mathrm{ch}\, X(G, H) \geq m$.*

We shall give a stronger version of this conjecture in purely graph theoretic terms (3.5).

Note that:

PROPOSITION 2.3. *Every group has a quasi-irreducible generating set.*

Proof. Let "$<$" denote a well-order relation on G and set

$$H = \{g: g \notin \langle h; h < g \rangle\} .$$

Obviously, H is a quasi-irreducible generating set.

In section 4, we shall prove

THEOREM 2.4. *Every (finite or infinite) solvable group has a Cayley graph whose chromatic number is at most 3.**

Let us remark that Conjecture 2.2 seems to be unsolved even for finite cyclic groups. It holds trivially, with $m = 3$, for elementary abelian groups. N. Biggs has asked whether (contrary to 2.2), for such complicated groups G as the (non-cyclic) simple ones,

---

* Independently, for finite groups, this has been proven by Saul Stahl (personal communication by A. T. White.)

ch $X(G, H)$ is necessarily large if H is a sufficiently large irreducible generating set. The answer is no: for n odd, A_{3n} has an irreducible set of $n + 1$ generators such that the resulting Cayley graph is at most 9-chromatic. (The generators are the cycles $(1, 2, 3), (4, 5, 6), \ldots, (3n - 2, 3n - 1, 3n)$ and $(3, 6, 9, \ldots, 3n)$.)

3. The proof of Theorem 2.1 and a graph theoretic conjecture.

DEFINITION 3.1. A graph X is called a no pied circuit graph if its edges can be colored such that

(i) each vertex is incident with at most two edges of any color;

(ii) each circuit contains at least two edges of the same color.

If we replace (ii) by

(iii) for any edge of any circuit there is another edge of the same color in that circuit, we call X a no lonely color graph.

PROPOSITION 3.2. For H a quasi irreducible generating set of G, $X(G, H)$ is a no pied circuit graph. If H is irreducible then $X(G, H)$ is a no lonely color graph.

Proof. We may assume that if $h, h^{-1} \in H$ then $h^2 = 1$. Let us take H as the set of colors.

We assign color h to the edge $[g, hg]$ $(g \in G, h \in H)$. The edges of color h incident with g are $[g, hg]$ and $[g, h^{-1}g]$ (they coincide iff $h^2 = 1$), hence (i) is satisfied. A circuit having colors $h_1, h_2, \ldots, h_k$ (in this order) means $h_1^{\pm 1} h_2^{\pm 1} \cdots h_k^{\pm 1} = 1$. Hence any lonely color can be expressed by the rest of the colors in this circuit. Thus if H is irreducible (iii) is satisfied. If H is quasi-irreducible and h_j is the greatest among $h_1, \ldots, h_k$ in the corresponding well-ordering of

H, then, by definition, h_j cannot be expressed by other colors of this circuit; hence $h_i = h_j$ for some $i \neq j$, $1 \leq i \leq k$, and (ii) is satisfied.

LEMMA 3.3. A no pied circuit graph does not contain any $K_{5,17}$ subgraph. A no lonely color graph does not contain any $K_{3,5}$.

Proof. Assume, to the contrary, that $K_{5,17}$ is a no pied circuit graph. Let $V(K_{5,17}) = A \cup B$, $|A| = 5$, $|B| = 17$, all edges going between A and B. Let $b \in B$. By 3.1 (i), there are at least 3 members a_1, a_2, a_3 of A such that the $[b, a_i]$ $(i = 1, 2, 3)$ have different colors. Let γ_i denote the color of $[b, a_i]$. By 3.1 (i) there are at most 5 edges joining a_i to $B \setminus \{b\}$ whose color belongs to $\{\gamma_1, \gamma_2, \gamma_3\}$; hence there is a vertex $b' \in B \setminus \{b\}$ such that none of the edges $[b, a_i]$ has any of the colors $\gamma_1, \gamma_2, \gamma_3$. On the other hand, there are at least two different ones among the colors of $[b', a_i]$, say $[b', a_1]$ and $[b', a_2]$ have different colors. But now, the circuit $b - a_1 - b' - a_2$ is pied (has 4 different colors), a contradiction.

The proof of the other statement of the lemma goes similarly; we omit it.

Observe that we used condition 3.1 (ii) (and (iii), resp.) for 4-circuits only.

Theorem 2.1 follows directly from Proposition 3.2, Lemma 3.3 and the following theorem of ERDŐS and HAJNAL:

THEOREM 3.1 (P ERDŐS and A. HAJNAL [8, Cor.5.6, p. 72]). If the chromatic number of a graph X is at least ω_1 then X contains a K_{m,ω_1} for any finite m.

Now we formulate a stronger version of Conjecture 2.2 which has the advantage for graph theorists that no groups occur in it.

CONJECTURE 3.5. *There is a finite* m *such that the chromatic number of any no pied circuit graph (no lonely color graph, resp.) does not exceed* m .

Of course, this would not be true if we assume 3.1 (ii) ((iii), resp.) for circuits of bounded lengths only, as there are finite graphs of arbitrarily large girth and chromatic number (P. ERDŐS, see [9, p. 54]).

4. Minimum chromatic number of Cayley graphs.

It is natural to define graph theoretic characteristics of a group as the extremum of the corresponding numbers for its Cayley-graphs (cf. [23, p. 69] for the definition of the *genus* of a group).

For a group G, let

$$\operatorname{ch} G = \min \{\operatorname{ch} X(G, H): \ H \text{ generates } G\} .$$

Thus Theorem 2.4 asserts:

THEOREM 4.1. *For* G *a solvable group,* $\operatorname{ch} G \leq 3$.

LEMMA 4.2. *For* $N \lhd G$, $N \neq G$ *we have*

$$\operatorname{ch} G \leq \operatorname{ch}(G/N) .$$

Proof. Let $\varphi: G \to G/N$ denote the natural epimorphism and let $X = X(G/N, H)$ be a Cayley-graph $(1 \notin H)$ with $\operatorname{ch} X = \operatorname{ch}(G/N)$. Let $Y = X(G, \varphi^{-1}(H))$. Clearly, $\varphi^{-1}(H)$ generates G. We assert that $\operatorname{ch} Y \leq \operatorname{ch} X$. For this observe that $[x, y] \in E(Y)$ iff $[\varphi(x), \varphi(y)] \in E(X)$; thus the coloring of X can be lifted to an appropriate coloring of Y,

Since the n-cycle is a Cayley-graph of the cyclic group Z_n, Lemma 4.2 makes Theorem 4.1 trivial for finite groups.

CONJECTURE 4.3. For any finite group, ch $G \leq 4$. (Probably, even ch $G \leq 3$ holds.)

Lemma 4.2 implies that it suffices to prove the conjecture for finite simple groups.

PROPOSITION 4.4 If every finite simple group can be generated by 2 of its members (as conjectured by group theorists, cf. [10]) then 4.3 is true. If one of the generators can always be chosen to be an involution than even ch $G \leq 3$ holds for any finite group G .

Proof. By 4.2 it suffices to prove that ch $G \leq 4$ (ch $G \leq 3$, resp.) for G simple. If $H = \{\alpha, \beta\}$ then $X(G, H)$ has degree 4(3, if α is an involution). Hence the proposition follows from Brooks' theorem, since, as H is irreducible, $X(G, H)$ does not contain K_4 (nor K_4^-, by 5.3 and 5.4), and a triangle in the second case would mean $\beta^3 = 1$ (5.3, 5.4). In this latter case $(\alpha^2 = \beta^3 = 1)$ let $N = (\alpha, \beta^{-1}\alpha\beta, \beta^{-2}\alpha\beta^2)$.

Clearly, $N \triangleleft G$, hence (G being simple) $N = G$. Thus G is generated by 3 involutions. The Cayley graph $X(G, \{\alpha, \beta^{-1}\alpha\beta, \beta^{-2}\alpha\beta^2\})$ has again degree 3 and it does not contain any triangle.

PROPOSITION 4.5. ch $S_n = 2$ $(n \geq 2)$, and ch $A_n = 3$ $(n \geq 3)$.

Proof. ch $A_n \leq 3$ follows, as in the above proof, from the observation that A_n can be generated by an involution and another permutation (for $n \geq 4$; $A_3 = Z_3$) . ch $S_n = 2$ and ch $A_n \neq 2$ both follow from the next observation:

PROPOSITION 4.6. For a (finite or infinite) group G , ch $G = 2$ iff G has a subgroup of index 2 .

Proof. Let $|G:N| = 2$; then by 4.2 we have

$$\text{ch } G \leq \text{ch } (G/N) = \text{ch } Z_2 = 2 .$$

Conversely, if $X(G , H)$ is a connected bipartite graph then the members having the same color as the unit form a subgroup of index 2.

We mention a corollary of 4.2 and 4.5.

COROLLARY 4.7. If G is the automorphism group of a finite planar graph, then $\text{ch } G \leq 3$.

Proof. By [3, Theorem 9.3, p. 72], the factors of the composition series of G are cyclic and alternating groups. Now we may apply 4.2 and 4.5.

The proof of 4.1 will be based on the following:

LEMMA 4.8. $\text{ch}(Z_{p^\infty}) = 3$.
(Z_{p^∞} denotes the additive group of the fractions a/p^k mod 1 ($a \in Z$, $k \in Z$, $k \geq 0$) for p a prime. Any proper subgroup of Z_{p^∞} is finite. This implies by 4.6 that $\text{ch}(Z_{p^\infty}) \geq 3$.)

Proof. Let $q = 4$ for $p = 2$ and $q = p$ otherwise. Let $H = \{1/q^k : k = 1 , 2 , \ldots\}$. We assert that $\text{ch } X(Z_{p^\infty} , H) = 3$. The colors will be called $\underline{0}$, $\underline{1}$ and $\underline{2}$.

Let us fix a 3-coloring $a_0 , a_1 , \ldots , a_{q-1}$ of a $(q - 1)$ - path such that $a_0 = a_{q-1} = \underline{0}$ (and $a_i \neq a_{i+1}$, of course). Let $H_n = \{1/q^k : k = 1 , \ldots , n\}$, $G_n = \langle H_n \rangle$. We shall successively 3-color the graphs $X_n = X(G_n , H_n)$; the color of x will be denoted by $c(x)$.

X_1 is a q-circuit; we 3-color it arbitrarily.

Let X_{n-1} be 3-colored. Let $x \in G_n \setminus G_{n-1}$; thus

$$x = y + s/q^k \quad \text{for some } s , \quad 1 \leq s \leq q - 1 \quad \text{and} \quad y \in G_{n-1} .$$

(s and y are unique.)

Set

$$c(x) \equiv c(y) + a_s \pmod 3 .$$

Assuming that c is a good coloring of X_{n-1}, $a_0 = a_{q-1} = \underline{0}$ and $a_i \neq a_{i+1}$ guarantee that $c(x) \neq c(x')$ if $x \equiv x' + 1/q^n \pmod 1$ $(x, x' \in V(X_n))$. If $x \equiv x' + 1/q^k \pmod 1$ for some k, $1 \leq k < n$, then $c(x) - c(x') \equiv c(y) = c(y') \pmod 3$; hence this is $\not\equiv 0$ as y and y' are neighbors in X_{n-1}. Thus, by induction, c is a good 3-coloring of $X(Z_{p^\infty}, H) = \bigcup_{n=1}^{\infty} X_n$.

Now we can turn to the proof of Theorem 4.1. As G is solvable, $G/G' = G_1$ is an abelian group, $|G_1| \geq 2$. Then G_1/pG_1 is an elementary abelian p-group for any prime p; thus it has Z_p as a factor group except if $G_1 = pG_1$. Thus either G has Z_p as a factor group for some p and ch $G \leq 3$ follows from 4.2, or G_1 is a divisible group, and hence the direct sum of copies of Z_{p^∞} $(p = 2, 3, 5, \ldots)$ and Q (the additive group of the rationals) (cf. FUCHS [11, Thm. 23.1]). Now $Q/Z \cong \sum_p Z_{p^\infty}$; hence in any of these cases, G_1 (and G) has some Z_{p^∞} as a factor group, proving ch $G \leq 3$ by 4.8 and 4.2.

5. Subgraphs of Cayley graphs

We start this section with proposing

PROBLEM 5.1. Which graphs Y are <u>avoidable</u> in the sense that every group has a Cayley graph not containing Y as a subgraph?

By 3.2 and 3.3, $K_{5,17}$ is avoidable (and, for finite groups, even $K_{3,5}$ is avoidable).

Recall a simple lemma [4, Proposition 5.6].

LEMMA 5.2 <u>Let</u> H <u>be a generating set of</u> G. <u>Then</u> H <u>contains a subset</u> K, <u>generating</u> G, <u>such that for any</u> $x_1, x_2, x_3 \in K$ <u>and</u> $\varepsilon_1, \varepsilon_2, \varepsilon_3 = \pm 1$,

$$x_1^{\epsilon_1} x_2^{\epsilon_2} x_3^{\epsilon_3} = 1$$

implies $x_1 = x_2 = x_3$, $x_1^3 = 1$.

Let us call such a generating set K 3-product-free.

5.3. A generating set K is reduced if it is quasi-irreducible and 2-product-free. By 2.3 and 5.2, every group has a reduced generating set. Note that an irreducible generating set is necessarily reduced.

PROPOSITION 5.4. For H a reduced generating set of G, $X(G, H)$ does not contain K_4^- (K_4 minus an edge). If $X(G, H)$ contains a triangle then one of the generators has order 3 .

Proof. Trivial.

COROLLARY 5.5. K_4^- is avoidable.

We now show that the n-cycles are simultaneously unavoidable.

PROPOSITION 5.6. The Cayley graphs of Z_3^n are pancyclic (they contain cycles of all possible lengths $3, 4, \ldots, 3^n$) .

Proof. It is readily checked that the Cartesian product (cf. [20]) of a 3-cycle by a path of length $k \geq 2$ is pancyclic. Now it follows by induction that the Cartesian product C_3^n of n copies of C_3 is pancyclic. But clearly, $X(Z_3^n, H) \cong C_3^n$ for any irreducible generating set H of Z_3^n .

As our proof of 5.5 shows, Problem 5.1 leads to questions concerning relations among generators of a group. One of the simplest problems of this kind is the following:

PROBLEM 5.7. Given an infinite group G, does it contain a subset H of power $|G|$ such that for any $x, y, z \in H$,

$$x^{-1} y = y^{-1} z \quad \text{implies} \quad x = z?$$

Partial results concerning 5.7 have been applied in [4] to prove that, given an infinite group G, there is a digraph whose automorphism group is transitive, and isomorphic to G.

Concerning 5.7, see also [12].

Let us close our remarks on problem 5.1 with:

CONJECTURE 5.8. A graph is avoidable (if and) only if it has an avoidable finite subgraph.

5.9. Finding the possible induced subgraphs of Cayley graphs is much easier than problem 5.1. H.W. LEVINSON has shown that any connected graph on n vertices may be embedded (as a induced subgraph) in some Cayley graph of a free group of rank $n - 1$ [14].

This result can be improved:

THEOREM 5.11. There is a mapping f of the class of cardinals into itself such that, given a graph X of order n, any group of order $\geq f(n)$ has a Cayley graph containing X as a induced subgraph. For n finite, $f(n)$ is finite; $f(\omega) = \omega$ (where ω is the cardinal number of the integers) and $f(\lambda) = \lambda$ for any weakly compact cardinal λ; and $f(\kappa) \leq (\sup_{\mu \leqslant \kappa} 2^{\mu})^{+}$ for any infinite κ. Under the generalized continuum hypothesis, $f(\kappa) \leq \kappa^{+}$. ($\kappa^{+}$ is the successor of cardinal κ.)

The proof of 5.11 (see [6]) is based on the partial answer to 5.7 given in [4, Lemma 3.2].

PROBLEM 5.12. Does 5.11 not hold with $f(\omega_1) = \omega_1$?

PROBLEM 5.13. Which graphs may be embedded (as induced subgraphs) in Cayley graphs of irreducible group presentations?

6. Remark. The investigation of contractions of Cayley graphs (and of graphs with prescribed automorphism groups) seems to be worth

mentioning here. For lack of space we have to content outselves with mentioning one of the results, which is a corollary to [1, Lemma 3] (cf. also [2]).

LEMMA 6.1. *For G a subgroup of the group H, any Cayley graph of H is contractible onto some Cayley graph of G.*

This lemma has a number of applications. It in turn implies the Nielsen-Schreier theorem saying that any subgroup of a free group is free, A.T. White's conjecture [23, problem 7-7] stating that the genus of a subgroup never exceeds that of the group, etc. For details see [5].

REFERENCES

1. L. BABAI, Groups of graphs on given surfaces, Acta Math. Acad. Sci. Hung. 24(1973), 215-221.
2. L. BABAI, Automorphism groups of graphs and edge-contraction, Discrete Math. 8(1974), 13-20.
3. L. BABAI, Automorphism groups of planar graphs II, Infinite and Finite Sets, Keszthely 1973 (A. Hajnal et al. eds.), North-Holland, Amsterdam 1975, 29-84.
4. L. BABAI, Infinite digraphs with given regular automorphism groups, J. Comb. Theory B, to appear.
5. L. BABAI, Some applications of graph contractions, to appear.
6. L. BABAI, Embedding graphs in Cayley graphs, to appear.
7. H. S. M. COXETER and W. O. J. MOSER, Generators and relations for discrete groups, Springer, Göttingen 1957.
8. P. ERDÖS and A. HAJNAL, On chromatic number of graphs and set-systems, Acta Math. Acad. Sci. Hung. 17(1966), 61-99.
9. P. ERDÖS and J. SPENCER, Probabilistic methods in combinatorics, Akadémiai Kiadó and Academic Press, Budapest-New York, 1974.
10. W. FEIT, The current situation in the theory of finite simple groups, Actes du Congrés Internat. des Math. I, Gauthier-Villars 1971, 55-93.
11. L FUCHS, Infinite abelian groups, I, Academic Press, N. Y. 1970.
12. J. L. HICKMAN and B. H NEUMANN, A question of Babai on groups, Bull. Austral. Math. Soc. 13(1975), 355-368.

13. H. A. JUNG, Ober den Zusammenhang von Graphen, mit Anwendungen auf symmetrische Graphen Math. Ann. 202(1973), 307-320.

14. H. W. LEVINSON, On the genera of graphs of group presentations III, J. Comb. Theory 13(1972), 209-302.

15. C. H C. LITTLE, D. D. GRANT and D. A. HOLTON, On defect-d matchings in graphs, Discrete Math. 13(1976), 41-54.

16. L. LOVASZ, Combinatorial Problems and Excercises, Akadémiai Kiadó-North-Holland, book to appear.

17. W. MADER, Über den Zusammenhang symmetrischer Graphen, Archiv der Math., 21(1970), 331-336.

18. W. MADER, Eine Eigenschaft der Atome endlicher Graphen, Archiv der Math. 22(1971), 333-336.

19. G. SABIDUSSI, On a class of fixed-point-free graphs, Proc. Amer. Math. Soc. 9(1958), 800-804.

20. G. SABIDUSSI, Graphs with given automorphism group and given graph theoretical properties, Canad. J. Math. 9(1957), 515-525.

21. M. E. WATKINS, Connectivity of transitive graphs, J. Comb. Theory 8(1970), 23-29.

22. M. E. WATKINS, On the action of non-abelian groups on graphs, J. Comb. Theory 11(1971), 95-104.

23. A. T. WHITE, Graphs, groups and surfaces, North-Holland, Amsterdam 1973.